DO NOW — RECALL

Lesson 125 · Assembly Language and the Little Man Com

Name: _____ Class: _____ Date: _____

1. What is the purpose of a function return value?

2. After pushing A then B onto a stack, which is popped first?

3. What is the output of a full adder with A=1, B=1 and carry-in=0?

4. Why is the program counter important in a stored-program computer?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It passes a computed result to the caller.

2. Model answer

B, since a stack uses LIFO.

3. Model answer

Sum 0, carry-out 1.

4. Model answer

It tracks the address of the next instruction to fetch.

By the end, you can…

• Explain the roles of the accumulator, program counter and numbered memory mailboxes in the LMC model.

• Use INP, OUT, LDA, STA, ADD, SUB, BRA, BRZ, BRP, HLT and DAT correctly in simple programs.

• Trace a short stored-program sequence and give the contents of relevant memory and registers.

Vocabulary — know these terms

LMC — Educational model of a stored-program accumulator computer.

accumulator — Register holding current working data and arithmetic results.

program counter — Register holding address of the next instruction to fetch.

mnemonic — Symbolic name representing a machine instruction.

NEW LEARNING

Read, annotate and explain

Mailboxes

Instructions and numeric data occupy addressed memory mailboxes. DAT reserves a data location. Programs execute the instructions addressed by the program counter.

Accumulator and PC

INP reads into ACC; LDA reads memory into ACC; STA writes ACC to memory. The PC normally advances after a fetch, unless a branch selects a new address.

Branches and stopping

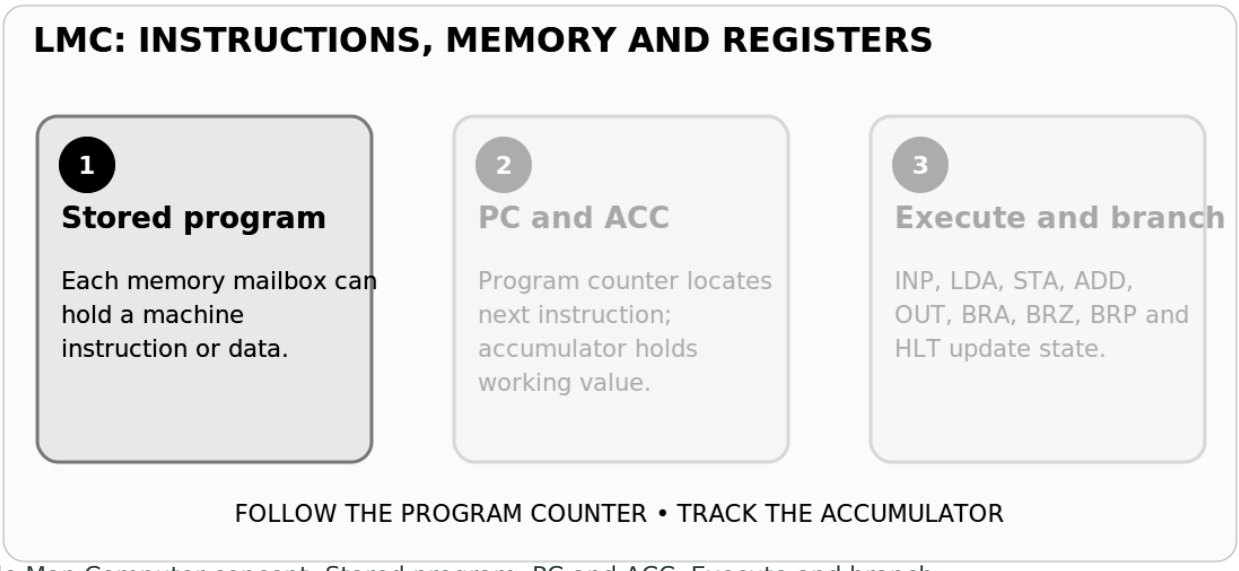
ADD and SUB update ACC; OUT prints ACC; BRA always branches; BRZ tests for zero; BRP tests a non-negative result in standard LMC; HLT ends the program.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

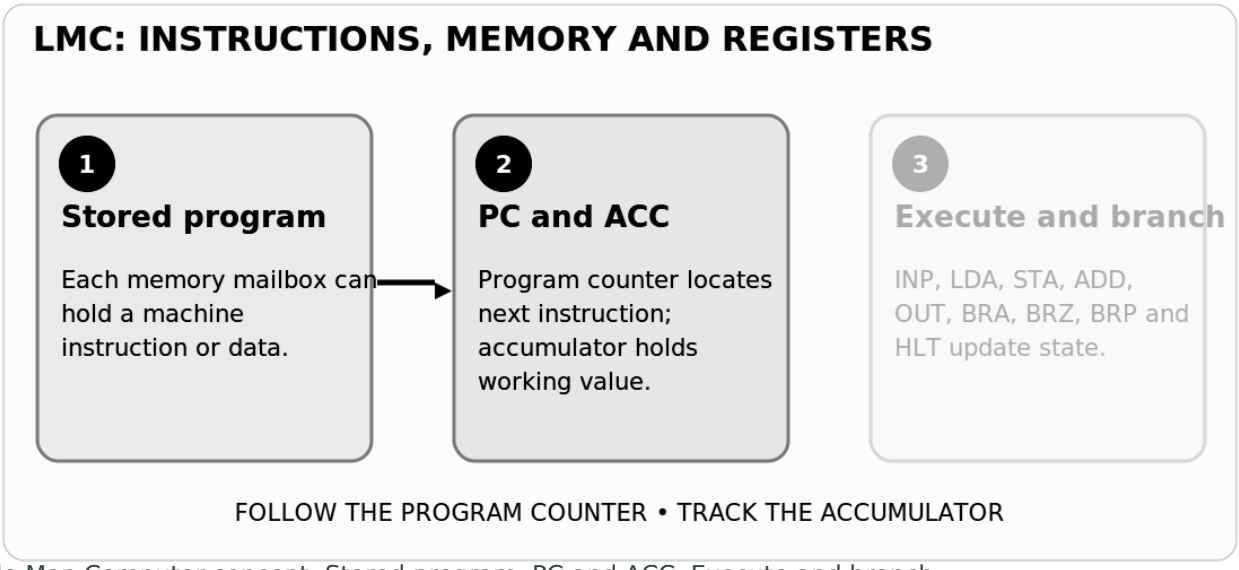
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Little Man Computer concept: Stored program, PC and ACC, Execute and branch

New Learning

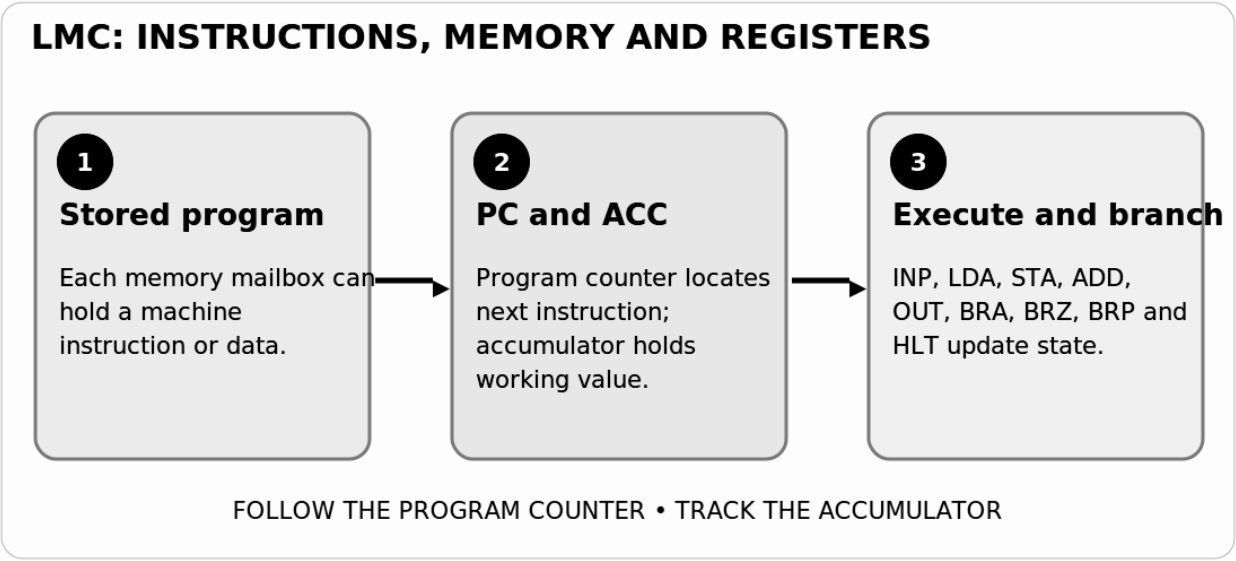


Little Man Computer concept: Stored program, PC and ACC, Execute and branch

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Little Man Computer concept: Stored program, PC and ACC, Execute and branch

I DO — WORKED EXAMPLE

Follow the worked method

Input is 7. Trace this program, including the accumulator, data mailbox and output.

Track each change

00 INP: ACC = 7.

01 STA 10: mailbox 10 becomes 7; ACC remains 7.

02 LDA 10: ACC = 7.

03 ADD 10: ACC = 14.

04 OUT: output is 14 .

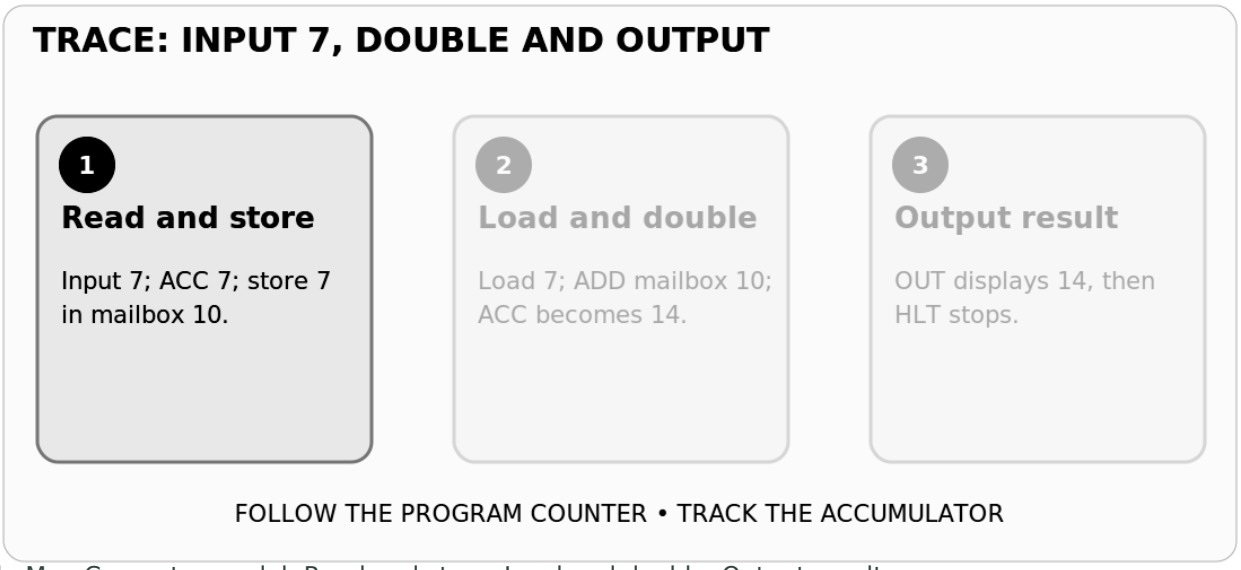
05 HLT: stop. Data mailbox 10 is separate from instruction addresses 00–05.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Input is 7. Trace this program, including the accumulator, data mailbox and output.

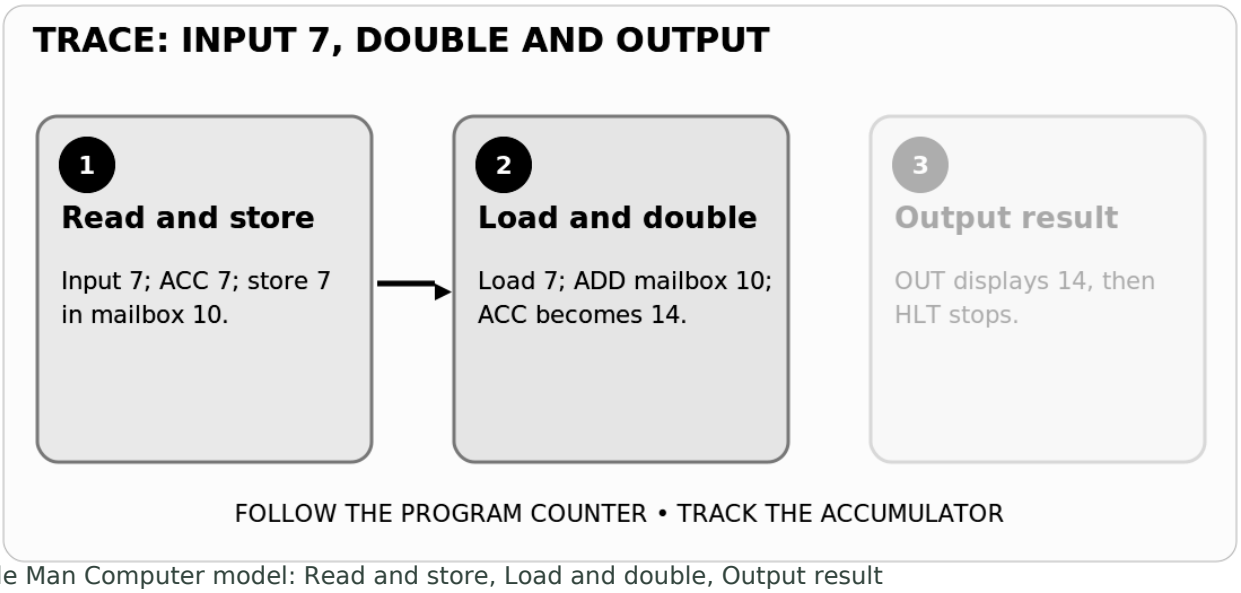
00 INP: ACC = 7.



Little Man Computer model: Read and store, Load and double, Output result

Input is 7. Trace this program, including the accumulator, data mailbox and output.

00 INP: ACC = 7.



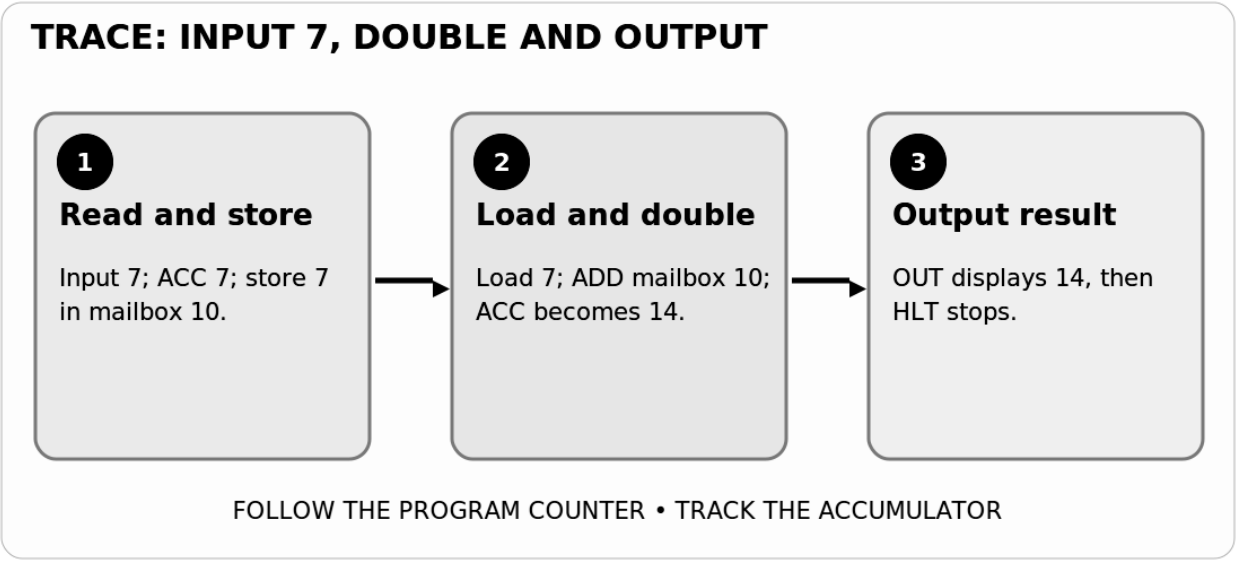
Little Man Computer model: Read and store, Load and double, Output result

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Input is 7. Trace this program, including the accumulator, data mailbox and output.

00 INP: ACC = 7.



Little Man Computer model: Read and store, Load and double, Output result

WE DO — GUIDED PRACTICE

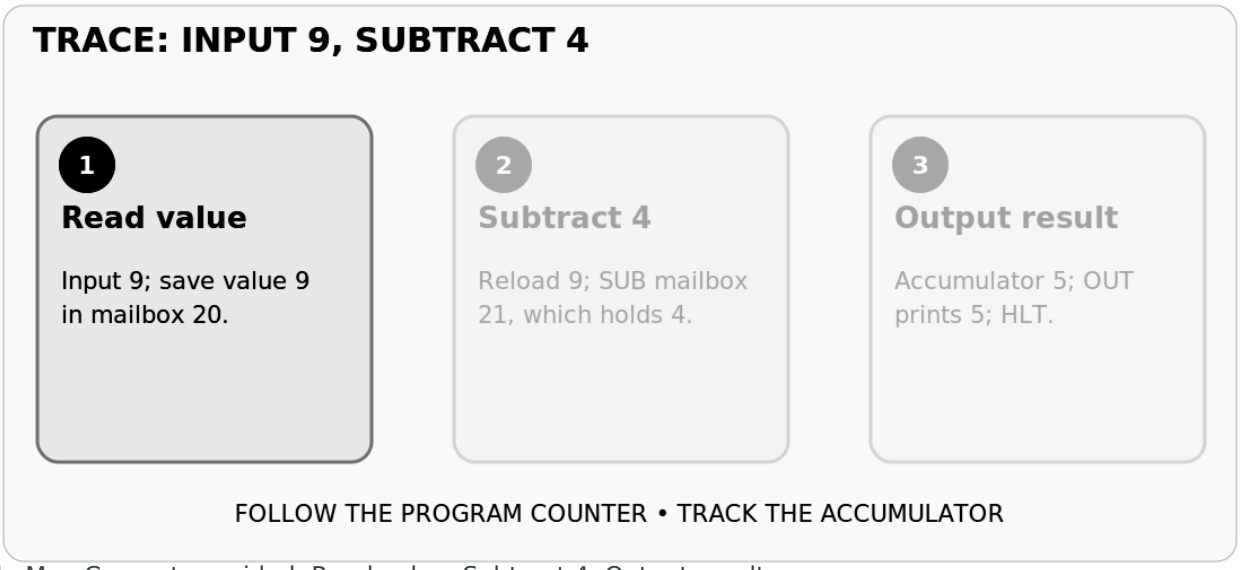
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Input is 9, and mailbox 21 contains 4. What is the output and why?

What are ACC and mailbox 20 after the first two instructions?



Little Man Computer guided: Read value, Subtract 4, Output result

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the registers What are ACC and mailbox 20 after the first two instructions?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain what the PC, accumulator and memory mailboxes represent in LMC. [6]

Question 2

Trace INP; STA 30; LDA 30; ADD 31; OUT; HLT, with input 5 and mailbox 31 containing 3. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write LMC instructions to read an integer and output it, using a BRZ branch when zero. Explain how you would test both paths. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In LMC, STA n loads the accumulator with the contents of mailbox n.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

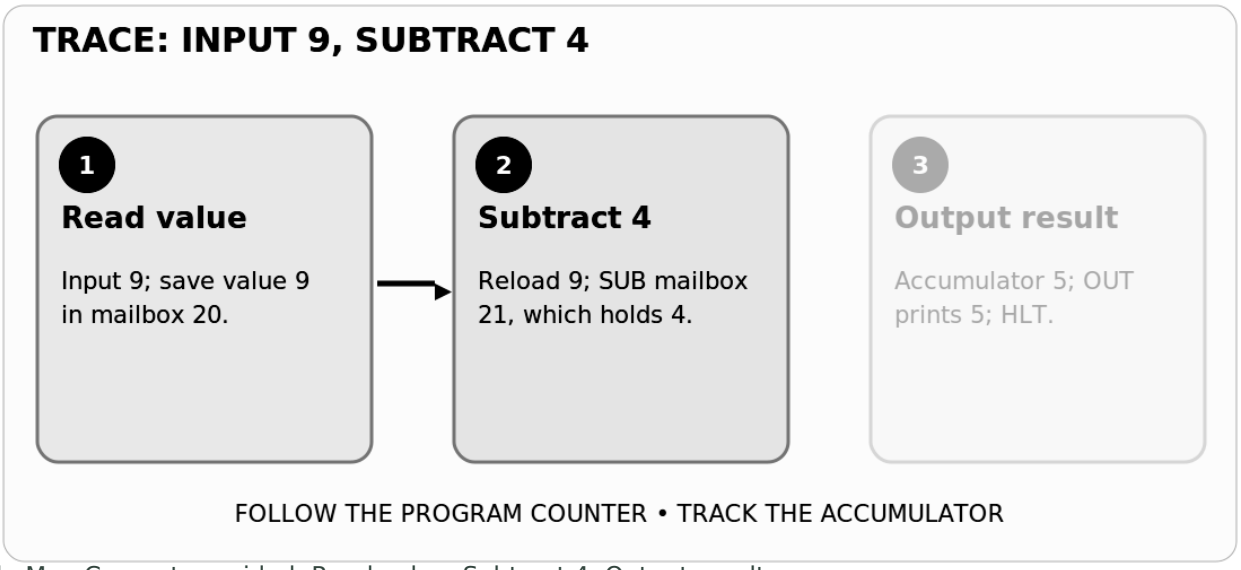
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Input is 9, and mailbox 21 contains 4. What is the output and why?

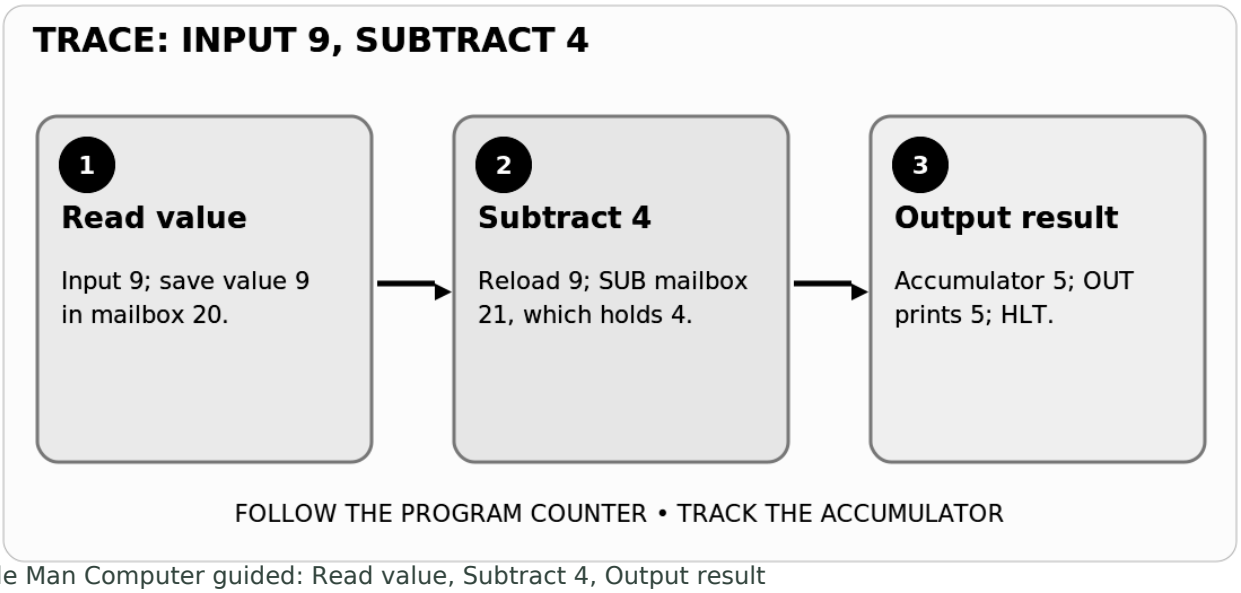
What are ACC and mailbox 20 after the first two instructions?



Little Man Computer guided: Read value, Subtract 4, Output result

Input is 9, and mailbox 21 contains 4. What is the output and why?

What are ACC and mailbox 20 after the first two instructions?



Little Man Computer guided: Read value, Subtract 4, Output result

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Calculate ACC

Follow LDA 20 and SUB 21. Use the stored value 9 and constant 4.

Completed trace

INP sets ACC to 9. STA 20 saves 9. LDA 20 reloads it. SUB 21 subtracts 4, leaving ACC = 5. OUT prints 5 ; HLT stops.

IMPROVE: Explain one correction to your own reasoning.